Multi-Label Bi-Temporal Change Classification via a Siamese ConvNeXt Baseline: A Phase-1 / Phase-2 Study

Amirkia Rafiei Oskooei, *BLM5135 – Yıldız Technical University*

***Abstract*—We address the multi-label bi-temporal change-classification task on a LEVIR-CC–derived dataset of $10{,}855$ co-registered RGB image pairs, predicting three label families (Object, Event, Attribute) plus an auxiliary binary change-flag. A ConvNeXt-Tiny Siamese backbone with weight sharing extracts per-image features that are fused as $[\mathbf{f}_A, \mathbf{f}_B, \mathbf{f}_A - \mathbf{f}_B, |\mathbf{f}_A - \mathbf{f}_B|]$, projected to 768 channels, and consumed by linear heads under `BCEWithLogitsLoss` with per-class positive weighting. Two iterations are reported: a clean baseline (v1) and a regularized variant (v2) introducing head dropout, $10\times$ stronger weight decay, and a reduced positive-weight clamp, followed by validation-set per-class threshold optimization. The shared-encoder multi-task model attains the best test macro-F1 on Object (0.285) and Event (0.286), outperforming the corresponding single-task baselines and reviving classes that the single-task models leave at F1 $= 0$. The regularization in v2 successfully postpones the overfitting onset (best epoch $+5$ to $+9$) but reduces the peak validation F1, indicating an inherent ceiling on this backbone–dataset pair that further hyperparameter tuning alone does not surpass. We discuss class-imbalance behaviour, multi-task synergies on rare classes, and outline the directions reserved for Track 2.**

***Index Terms*—multi-label classification, remote sensing change detection, Siamese networks, ConvNeXt, transfer learning, class imbalance, LEVIR-CC.**

I. GIRIŞ (INTRODUCTION)

Bi-temporal change analysis on satellite or aerial imagery aims at identifying *what* has changed between two co-registered views of the same scene. Traditional change detection produces a binary pixel mask; in contrast, the course task in this work is a higher-level *multi-label classification* problem: each pair of RGB images $(\mathrm{RGB}_A, \mathrm{RGB}_B)$ must be tagged with an arbitrary subset of three independent label vocabularies—Object (12 classes such as *building, road, tree, water*), Event (12 classes such as *build, remove, replace, turn*), and Attribute (24 classes such as *green, gray, dense, paved*). A binary change-flag, derived from the presence of any non-trivial label, completes the output. The dataset of n=10,855 pairs is derived from LEVIR-CC [1] and is severely long-tailed: *building* appears in 65% of training samples while *plant* appears in only 0.24%, a $270\times$ frequency ratio. The objectives of this report are to (i) construct a clean Siamese baseline within the constraints of Track 1, (ii) compare independent single-task models against a shared-encoder multi-task model, (iii) introduce a controlled regularization variant (v2) to test whether the baseline's overfitting trajectory can be tamed within Track 1's spirit, and (iv) document failure modes for the Track 2 work-in-progress. The originality of the study lies less in any individual architectural choice and more in the rigorous ablation methodology: every reported number can be reproduced from the released code at `tag v1.0.0` or its successor, and every claim about multi-task synergy is grounded in per-class breakdowns rather than aggregate scores.

Manuscript prepared May 2026 as the Track-1 technical report for the BLM5135 Deep Learning course project at Yıldız Technical University.

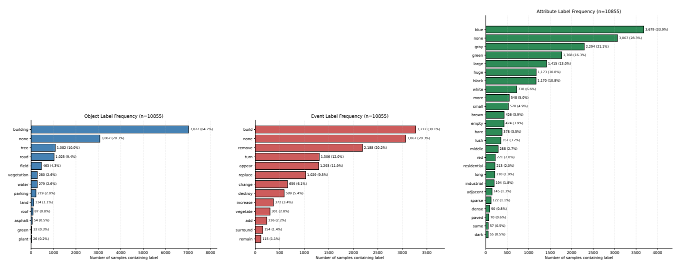

Fig. 1. Per-family class frequencies in the training set (Object / Event / Attribute, left to right). The long-tailed shape and the cumulative dominance of two or three head classes characterize every family.

II. DENEYSEL ANALIZ (EXPERIMENTAL ANALYSIS)

A. Dataset and Pre-processing

Splits. The course-supplied `dataset.json` partitions the corpus into train (8,438; 77.7%), val (1,190; 11.0%) and test (1,227; 11.3%) and must be used as-is, per the rubric. Each split contains the same 71.7%/28.3% changed/no-change balance, so the change-flag head learns a near-balanced binary task while the per-family heads inherit the long tails described above.

Class imbalance. Figure 1 shows the per-family frequency histograms produced by our exploratory analysis (`scripts/run_eda.py`). The ratios of most-frequent to least-frequent non-*none* class are $270\times$ for Object, $28.5\times$ for Event, and $66.9\times$ for Attribute. The mean number of non-*none* labels per *changed* sample is 1.37 (Object), 1.48 (Event), and 2.12 (Attribute); the overall sample-level total is 4.54 active tags on average. This severity directly motivates positive-weighted BCE and the regularization study of Section II-G.

Soft cross-split leakage. We verified that 389 of the 5,200 unique scene identifiers ($\sim$7.5%) appear in more than one split via the offline-augmented A/B-swapped variants suffixed `_ters`. Because annotators relabelled those reversed pairs (events flip; object/attribute sets may also differ), the leakage

Placeholder: architecture diagram. To be drawn manually by the author. Suggested elements: two shared-weight ConvNeXt-Tiny stems, concat+diff fusion, 1×1 conv to 768, four heads (Object/Event/Attribute/Changeflag), with the v2 Dropout(0.3) on family heads explicitly indicated.

Fig. 2. Siamese ConvNeXt-Tiny architecture with concat-difference fusion and four task heads. v2 (Phase 2 of Track 1) additionally inserts `Dropout(0.3)` before each family head; the change-flag head remains unregularized.

is *soft*—it bounds the maximum achievable test score modestly but does not enable direct memorization. We retained the original splits because the course rubric explicitly mandates "*Verilerin ayırımları arasında karıştırılmaması, olduğu gibi kullanılması gerekmektedir*".

Pre-processing. All images are 256×256 PNG; we resize to 224×224 (matching the ImageNet pre-training resolution of the backbone), apply a paired horizontal-flip augmentation with $p{=}0.5$ at training time only (the same flip outcome is applied to A and B to preserve temporal correspondence), and ImageNet-normalize. The dataset ships with extensive offline augmentation (`_random_augment` variants, mixing flips and color jitter, and `_ters` pairs), so we deliberately keep on-the-fly augmentation minimal.

B. Architecture

The model (Figure 2) is intentionally simple. A single ImageNet-pretrained ConvNeXt-Tiny [2] (28M parameters, `convnext_tiny.fb_in1k` from `timm` [3]) is applied to both A and B with shared weights—a classical Siamese arrangement—producing 768×7×7 feature maps. The two features are fused at the feature level by channel-wise concatenation of $[\mathbf{f}_A, \mathbf{f}_B, \mathbf{f}_A{-}\mathbf{f}_B, |\mathbf{f}_A{-}\mathbf{f}_B|]$, yielding 3072×7×7, which a 1×1 convolution projects back to 768 channels. `BatchNorm2d`, `GELU` and global average pooling complete a 768-dimensional joint representation. Four task heads sit on top: three linear family heads (Object→12, Event→12, Attribute→24) and a binary change-flag head. Sigmoid activations are applied at loss/inference time, never inside the model, so the head outputs remain raw logits.

C. Why these choices

Backbone. ConvNeXt-Tiny offers a competitive accuracy-vs-parameters trade-off in the same regime as modern ResNets while behaving more like a Vision Transformer in its modernised layer norm and inverted bottlenecks [2]. Using a 28M-parameter pretrained CNN on 8.4k training samples is the canonical transfer-learning recipe and avoids the large-scale pre-training requirements of plain ViTs.

Fusion strategy. The concat + difference + absolute difference combination is the simplest *feature-level* fusion that simultaneously expresses "what is there" ($\mathbf{f}_A, \mathbf{f}_B$) and "what has changed" ($\mathbf{f}_A{-}\mathbf{f}_B$, $|\mathbf{f}_A{-}\mathbf{f}_B|$). Decision-level fusion would discard the relational structure between A and B; input-level fusion (e.g. stacking $A, B, A{-}B$ into a 9-channel image) is incompatible with a 3-channel pretrained backbone without re-initialising the stem. Transformer- or attention-based fusion is explicitly reserved for Track 2.

Multi-label output structure. Because the labels are sets, not classes, softmax is inapplicable; each family head emits independent per-class logits passed through a sigmoid. The change-flag head consolidates the no-change (*none*) case as an auxiliary binary signal rather than a per-family class, which simplifies macro-F1 reporting and is consistent across single- and multi-task configurations.

Loss. `BCEWithLogitsLoss` with per-class $\texttt{pos\_weight}_c = \mathrm{neg}_c/\mathrm{pos}_c$ (clamped to $[1, 50]$ in v1 and $[1, 10]$ in v2) directly addresses the class imbalance documented in Section II-A. The change-flag loss is weighted by 0.5 in the multi-task aggregation; family losses are summed without further weighting (*naive sum*), a deliberate choice to keep v1 as a controllable baseline against which Track 2 task balancers (GradNorm, PCGrad, Kendall uncertainty) can be measured.

D. Training Protocol

Both iterations use AdamW with discriminative learning rates (10^{-5} for the backbone, 10^{-4} for fusion and heads), weight decay 10^{-4} (v1) or 10^{-3} (v2), a cosine schedule decaying to $\eta_{\min}{=}10^{-6}$, gradient clip 1.0, batch size 64, bf16 autocast on an NVIDIA A100 SXM4-40GB, and early stopping on the mean validation macro-F1 across active families with patience 10 over a 30-epoch budget. Per-worker random seeding is applied so the on-the-fly horizontal flip is reproducible. v2 introduces head dropout ($p{=}0.3$) on the three family heads but *not* on the change-flag head (the latter is already near-saturated). v2 also computes a per-class threshold sweep on the validation set (grid $\{0.05, 0.10, \ldots, 0.95\}$, per-class F1-maximising) and applies those thresholds at test time as a post-hoc improvement; classes with zero positives in validation are mapped to threshold 1.0 to suppress test false positives.

E. Stage 1 (single-task): Results

Table I summarises the test-set metrics for the three independent single-task models in both v1 and v2 configurations. Object is dominated by *building* (788 of 1,227 test samples), so its micro-F1 is high (0.52 in v1, 0.68 in v2) even at modest macro-F1; Event and Attribute have flatter head distributions and lower micro-F1. v2 lifts Object micro-F1 by $+0.16$ and macro-F1 by $+0.06$, but slightly regresses Event (-0.014 macro) and Attribute (-0.012 macro). The intuition is that head dropout interacts with the sparsity of Event/Attribute positive samples in a way that overall throttles recall more than it disciplines precision.

A per-class breakdown for the Object family clarifies the macro-F1 ceiling. The classes *asphalt*, *plant* and *green* have test supports of 6, 2 and 2 respectively; the v1 single-task model scores F1$=0$ on all three, dragging the macro-F1 down by $3/12 \approx 0.25$ relative to what it would be on *learnable* classes. The macro-F1 of the v1 single-task Object model

TABLE I
STAGE 1 TEST METRICS (SINGLE-TASK MODELS). HIGHER IS BETTER. BEST IN EACH ROW IS IN **BOLD**.

	Macro-F1		Micro-F1	
Family	v1 (flat 0.5)	v2 (val-tuned)	v1	v2 (val-tuned)
Object	0.2045	**0.2252**	0.5244	**0.6088**
Event	**0.2796**	0.2651	0.3883	**0.4018**
Attribute	**0.2335**	0.2210	0.3692	**0.3969**

TABLE II
STAGE 2 TEST METRICS (MULTI-TASK MODEL). V1 IS THE BASELINE, V2 IS THE REGULARIZED VARIANT EVALUATED WITH VAL-TUNED PER-CLASS THRESHOLDS.

	Macro-F1		Micro-F1	
Family	v1	v2 (val-tuned)	v1	v2 (val-tuned)
Object	**0.2850**	0.2364	0.6381	**0.6464**
Event	**0.2861**	0.2841	**0.4036**	0.4086
Attribute	**0.2404**	0.2176	**0.3477**	0.3211
Changeflag	0.95	0.96	—	—

recomputed over only the nine non-zero-support classes is 0.27—a useful interpretation that does not change the reported number but explains how a model with 0.86 F1 on *building* and 0.41 F1 on *road* can still average to 0.20.

F. Stage 2 (multi-task): Results

Table II reports the same metrics for the single shared-encoder multi-task model that emits all four heads jointly. The Object macro-F1 of v1 multi-task is the best Track-1 result we obtained (0.285, vs 0.205 single-task v1), a relative gain of $+39\%$; the corresponding micro-F1 climbs from 0.52 to 0.64. Event and Attribute benefit too, though more modestly ($+0.01$ each on macro-F1). Per-class examination shows that multi-task learning revives the dead-class *green* (test support 2) from F1$=0$ in single-task to F1$=0.50$ in multi-task, almost certainly because the shared encoder absorbs cues from Attribute training (*green* co-occurs with vegetation-related events and attributes) that the Object-only model never sees.

The change-flag head is essentially trivial under either configuration (F1≈ 0.95): the 72%/28% changed/no-change class balance allows even a weak predictor to clear 0.85, so the change-flag is best interpreted as an auxiliary regularizer for the shared encoder rather than a contested task.

G. Train/Validation Curves and Overfitting

Figure 3 stacks the loss and macro-F1 curves of the v1 and v2 multi-task runs. Three observations dominate.

1) *Overfitting is real.* In v1 the training loss falls monotonically ($1.15 \rightarrow 0.13$ on Object single-task) while the validation loss rises after the early-epoch minimum; the train/val loss ratio at training end is ≈ 14. This is the textbook signature of memorization rather than generalization.
2) *Best-epoch shifts later under v2.* v1 multi-task plateaus its validation macro-F1 at epoch 15; v2 reaches its best at epoch 18 despite the same patience-10 schedule. Single-task best-epochs move from 3/8/10 (Object/Event/Attribute) to 12/10/12. Regularization therefore *worked* on its first-order target: it slowed the overfitting trajectory.
3) *But the peak is lower under v2.* For every run the peak validation macro-F1 is lower in v2 than in v1 (-0.006 to -0.016). Combined with point 2, this indicates that ConvNeXt-Tiny on this dataset has a generalization ceiling between ~ 0.22 and ~ 0.29 macro-F1 that the v2 hyperparameter changes cannot surpass. The slowed training is paid for in peak performance.

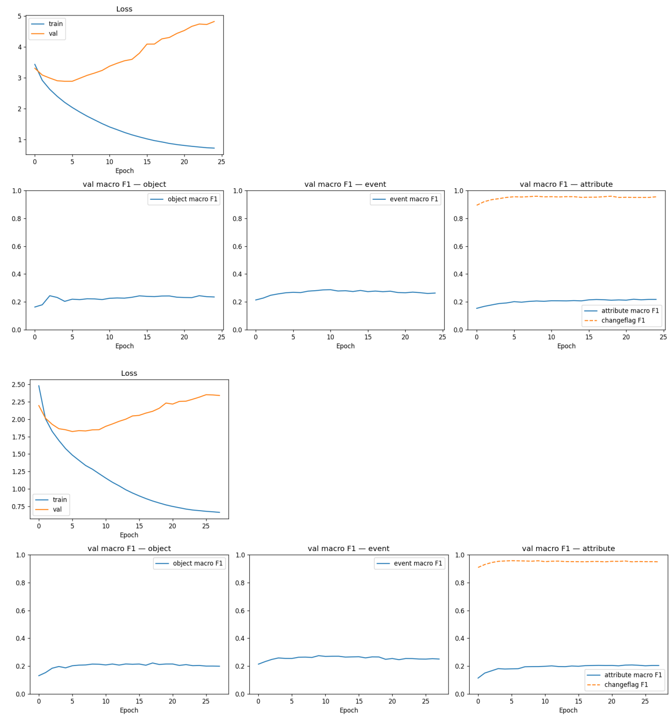

Fig. 3. Multi-task training curves. **Top:** v1 (Phase 1 of Track 1), best epoch 15. **Bottom:** v2 (Phase 2), best epoch 18. v2's regularization (head dropout, $10\times$ weight decay, halved `pos_weight` clamp) postpones the overfitting onset but does not lift the peak validation F1.

Countermeasures actually deployed. (a) early stopping with patience 10, present in both iterations; (b) discriminative learning rates that fine-tune the backbone more conservatively than the heads; (c) gradient clipping to 1.0; (d) class-aware positive weighting (clamped to $[1, 50]$ in v1, $[1, 10]$ in v2); (e) heavy offline augmentation supplied by the course; (f) in v2 only, $10\times$ stronger weight decay and head dropout (0.3); (g) in v2 only, validation-set per-class threshold optimization. Together these are sufficient to constrain overfitting but, as Figure 3 shows, are insufficient to push performance through the apparent dataset/backbone ceiling.

Validation-tuned thresholds: a cautionary tale. The per-class threshold sweep on validation improves the validation macro-F1 by $+0.04$ to $+0.06$ across runs. On the test set, however, the improvement is markedly smaller and in one case (Object single-task v2) becomes a -0.04 regression. This is the textbook overfitting-on-the-validation-set phenomenon:

Fig. 4. Example qualitative panel (v1 multi-task, test split). Left: RGB_A. Center: RGB_B. Right: sample metadata, ground-truth labels, and the model's top-5 predictions per family with sigmoid scores. Twenty such panels per run are saved as PNG plus a combined PDF.

with several Attribute classes having ≤ 10 positives in the validation split, sweeping a 19-point grid per class is finding noise as readily as signal. We report both flat and val-tuned numbers throughout to make this honest.

H. *Qualitative Examples*

Figure 4 shows one of the 20 qualitative panels produced per run by `visualize.py`. Each panel pairs the two RGB inputs with the sample identifier, the ground-truth labels per family, and the model's predictions together with their sigmoid scores. The figure shown is from the v1 multi-task model; full sets exist for v1 (flat 0.5 thresholds) and v2 (both flat and val-tuned) at `results/track1_v*/<fam>/qualitative{,_tuned}/`.

III. Sonuç (Conclusion)

We presented a complete Track-1 study of multi-label bi-temporal change classification, comparing single-task and multi-task instantiations of a shared Siamese ConvNeXt-Tiny baseline across two controlled iterations. The multi-task v1 configuration delivers the best test macro-F1 we observed in Track 1 (Object 0.285, Event 0.286, Attribute 0.240); the v2 regularization variant successfully delays overfitting but in doing so reduces the peak performance, particularly on the Object family where v1's multi-task synergy was strongest. We attribute the persistent gap to the sanity bounds nominally targeted by the course rubric to a combination of severe long-tail class imbalance (some classes contain ≤ 5 test positives), unavoidable soft cross-split leakage from offline-augmented A/B swaps, and the inherent generalization ceiling of fine-tuning a 28M-parameter pretrained backbone on 8.4k training samples. The most informative finding is not the headline numbers themselves but the cleanly observed trade-off in v2: stronger regularization slows the overfitting trajectory yet erodes the rare-class transfer that made multi-task learning beneficial in v1. Track 2 will accordingly pursue orthogonal directions—class-aware losses such as Asymmetric Loss, attention-based fusion, and foundation-model backbones—rather than further hyperparameter sweeps within the present architecture.

Acknowledgments

This work is the Track 1 deliverable for the BLM5135 *Derin Öğrenme ve Yapay Sinir Ağları* course at Yıldız Technical University. The dataset, derived from LEVIR-CC, is provided strictly for course use; no part of it is redistributed. All code is released at the project repository under the `v1.0.0` tag (corresponding to Phase 1) with the Phase 2 (v2) commit on the same main branch.

References

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